Justin Gnananadchtheram

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EDUCATION

University of Toronto **Sep. 2021 – April 2026**

Honours Bachelor of Science, Computer Science & Mathematics (Double Major) + PEY Co-op

Relevant Coursework: Artificial Intelligence, Machine Learning, Computer Security, Data Structures, Algorithms, Software Design, Cryptography, Operating Systems, Databases

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, Bash, HTML, CSS, TypeScript/JavaScript
Developer Tools: Git/GitHub/Bitbucket, PostgreSQL, Linux, Jenkins, Kali Linux
Technologies/Frameworks: NumPy, scikit-learn, JavaFX, React, Framer Motion, Mantine UI

EXPERIENCE

SPIN VFX **May 2024 – May 2025**

Software Developer Intern *Toronto, ON*

- Automated VFX scheduling via **Python event listeners** integrated with **Autodesk Flow Production**, enabling near real-time propagation of task and assignment changes across multiple teams.
- Built a sprint planning tool using **Python Streamlit** and the **Jira REST API**, generating schedules for future sprints, reducing planning time by **50%**, and providing a high level overview of developers' tasks and targets.
- Modularized legacy **Python** scripts leveraging **rez package manager**, eliminating an efficiency bottleneck between development and releases for essential daily cron jobs, reducing error triage time from hours to minutes.
- Refactored legacy internal **Python packages** introducing robust API error handling, significantly reducing timeouts due to API errors on daily cron jobs.
- Eliminated the need for manual schedule updates using task-schedule synchronization, saving VFX supervisors **5+ hours per week** while reducing update latency from hours to seconds.

PROJECTS

City Prediction ML Model | Python, NumPy, scikit-learn

- In a team of four, trained a machine learning model to **predict cities** based on student survey responses about architecture, culture, temperature, and word associations.
- Engineered data representations including numerical scaling, indicators, **bag-of-words** text features, and outlier clipping to reduce noise in the data and improve model robustness.
- Researched and trained a variety of models including **k-NN, Random Forest, and Neural Network**, evaluating their effectiveness by comparing test results, confusion matrices, rates of convergence, and more.
- Tuned a **multi-layer perceptron** using mini-batch gradient descent, optimizing learning rate, batch size, and regularization to improve accuracy.

Website Builder | React, TypeScript, Mantine UI, Framer Motion

- Architected a config-driven web platform using **React, TypeScript, Mantine UI,** and **Framer Motion**, enabling rapid generation of fully customized client websites from structured configuration files.
- Designed a **modular component and layout system** separating application logic from client content and branding, improving maintainability, scalability, and reducing expected website deployment time to day.
- Developed reusable, **responsive UI** components with typed configuration schemes, establishing a scalable foundation for **multi-client deployments** and consistent design standards.

Computer Security | C, Python, Kali Linux

- Exploited vulnerabilities in software from **OWASP** using buffer overruns, race conditions, SQL injections, fuzzing, cross site scripting, and password hash cracking.
- Conducted **forensic computing** on a compromised virtual machine, identifying attack vectors and vulnerabilities.
- Extracted malicious communications through examining captured packets using **Wireshark** on **Kali Linux**.

BOGGLE 1.1 | Java, JavaFX

- Developed a full-featured **Boggle game** in a team of four, featuring an interactive GUI, multiple difficulty modes, and language and accessibility settings.
- Created **UML diagrams** to assist in implementing **object oriented software design patterns** to improve code structure, readability, and to provide options for future additions.
- Collaborated using **GitHub** while following an **Agile** workflow to manage tasks and iterative development.